SRUTHI KORLAKUNTA

Automation Engineer

Details

San Jose, California, United States
6692604849
sruthi.korlakunta@gmail.com

Links

Sruthi's LinkedIn
Sruthi's Github

SKILLS

AI & MACHINE LEARNING
- Claude code
- Image Processing
- Defect Detection
- Random Forest Algorithm
- Statistical Analysis
- Time Series Forecast
- Generative Adversarial Networks
- Convolutional Neural Networks

PROCESS AUTOMATION
- UI Path
- PowerApps
- Python

DATA ANALYSIS
- Python
- SQL
- PowerBI
- KNIME
- Excel for Data analytics
- Tableau
- Dashboard Design
- Data Storytelling

CLOUD TECHNOLOGIES
- AWS Athena
- AWS Lambda
- AWS QuickSight

LANGUAGES
- German (Fluent, C1)
- English (Fluent)

Profile

Automotive Data Scientist with 8+ years of experience at BMW, Audi, and VW, specializing in the link between raw data and production-ready insights - such as sensor logs and perception training pipelines. I also architected scalable AWS ecosystems (Athena, Lambda, Glue, QuickSight) to automate high-dimensional data mining.

With 4 years of AI research in Computer Vision, I specialize in GAN-based synthetic data generation and network pruning to optimize model performance without losing "corner case" knowledge. I leverage OpenCV and statistical root-cause analysis to identify system failures, ensuring that perception models are resilient against edge-case anomalies.

Known consistently to self-teach and own the end-to-end implementation of new tools and technologies. I am an early adopter of LLM-augmented engineering, familiar with Claude code, Google AI studio, and Copilot into my workflow.

EDUCATION

M.Sc. Information and Automation Engineering, University of Bremen, Germany
(Focus: Artificial Intelligence, Machine Learning, Robotics, and Industrial Automation)
APRIL 2015 — APRIL 2018

B. Sc. Electrical and Electronics Engineering, Osmania University, India
APRIL 2009 — APRIL 2013

Certificates:
Claude Code in Action, Anthropic Academy, 2026
AI Fluency, Anthropic Academy 2026
Claude 101, Anthropic Academy 2026
UI Path Developer Licence, 2025
Currently pursuing: Introduction to LangGraph

Baltic Summer School – Cybersecurity and Cryptography, Estonia 2015

AI PROJECTS

NOVEMBER 2025 – MARCH 2026

- Built a Job-finder app on Google AI Studio using Gemini as a part of a Kaggle.com contest for AI enthusiasts – Github link
- Built an AI-powered React component generator with live preview (User describes a UI component in plain text and lets Claude write the code and render it instantly) using Claude code and VS code – Github link
- Built a Custom chatbot using Claude API – Github link

PROFESSIONAL EXPERIENCE

Digitization and Process Automation Lead at BMW Motorcycles (BMW AG), Berlin

OCTOBER 2024 — OCTOBER 2025

- Automated large-scale **data mining pipelines** from various sources, including autonomous tugger vehicle sensor logs, SAP ERP workflow data with AWS, Python and SQL, transitioning from manual processes to 100% cloud-native solutions that reduced reporting latency to zero.
- Leveraged statistical analytics to identify anomalies in supplier packaging, successfully rectifying root causes for 90% of identified line-stop triggers. Technologies used: Python, Scikit-learn.
- Automated several supply chain operations as bolt-on solution on top of SAP ERP, using UIPath and Python. For example, I built a detection system that periodically checks for stock in multiple storage locations and automatically diverts oncoming material to an available storage location.
- Led a project to reduce assembly line downtime by 80% by developing a predictive material availability model using machine learning and adjusting system metrics to ensure "right place, right time" delivery. Technologies used: Python, Scikit-learn.
- Mitigated urgent order backlogs by providing real-time visibility to site leaders on potential bottlenecks. I built a python automation that extracts relevant data from SAP's Urgent Order tracking system and built a real-time monitoring dashboard for site leader tracking. Technologies used: Python, AWS S3, Athena, Lambda, and QuickSight.

Lead Sales Data Engineer, Sales Coordination (Europe) at Audi AG, Ingolstadt

OCT 2019 — SEPTEMBER 2024

- Headed and executed a total pivot from purely excel-based reporting into Python, SQL and Power BI solutions. Automated Market Intelligence by architecting an end-to-end data pipeline using Python and SQL
- Removed sales planning obstacles for the Sales Managers by innovating reporting structures and re-coding data architecture that allowed comparisons of Audi models against competitors with high granularity, thus empowered 31 European markets with tools to identify market-share leaks and a identifying of true competitive threat.
- Lead cross-functional synchronization between Finance, Volume Planning, and Supply Chain that harmonized sales objectives with realistic supply chain capacity, significantly reducing supply-demand gaps and ensuring that financial forecasts were backed by physical inventory.
- Orchestrated a massive-scale IT compliance recovery, managing a user access inventory across 200 departments to meet stringent audit requirements and Led the logistics team through a 100% successful IT audit, serving as the primary point of contact for auditors and resolving all high-priority security findings.

Graduate Trainee, Perception and Machine Learning –Volkswagen AG, Wolfsburg

APRIL 2018 — OCTOBER 2019

- Research initiative at the Electronics Research Lab (California): optimizing **Object Detection and Classification** networks for autonomous vehicle deployment. Successfully utilized **network pruning** to reduce model footprint and training latency while maintaining rigorous **Data Quality Objectives (DQO)** to ensure zero degradation in 'corner case' performance.

  Applied the **Kullback-Leibler (KL) Divergence test** to measure the distance between the output probability distributions of the two models, ensuring the 'compressed' logic mirrored the original intent.
- Research project at Audi, Ingolstadt: Developed a high-fidelity **data mining pipeline** at Audi (Germany) to infer vehicle health metrics from **raw CAN-bus sensor logs**. Engineered features for SVM and Random Forest models to identify tire wear patterns without dedicated hardware, successfully extracting signal from noise in large-scale, low-level vehicle data.

Intern at Robert Bosch GmBH

APRIL 2017 — JULY 2017

- Engineered a production-level classifier for **high-dimensional sensor data,** utilizing **Random Forest classifier** to categorize complex vehicle states. This system enhanced root-cause analysis (RCA) protocols by identifying correlations between sensor telemetry and mechanical defects (False fuel types that were injected), improving fleet-wide diagnostics.

Master Thesis

- Developed a **computer vision** pipeline to synthesize realistic traffic environments by leveraging **GANs conditioned on bounding box** and **semantic segmented datasets**. Utilizing the KITTI and Cityscapes datasets, I engineered a GAN framework that generated high-fidelity ground truth samples for training perception models.

Student Researcher at BIBA

AUGUST 2016 — APRIL 2017

- Developed an end-to-end computer vision pipeline for **micro-surface defect detection** using **CNNs and image processing techniques**.
- Augmented datasets (because of rare occurrences of such defects in real life) by synthesizing defective datasets through custom augmentation algorithms using OpenCV.

Student Consultant at KPMG, Germany

AUGUST 2015 — AUGUST 2016

- Performed Enterprise IT Risk Management Analytics to evaluate system integrity and ensure rigorous adherence to financial reporting and regulatory ITGC standards, specifically in the SAP environment.
- Led complex data-driven audits to identify operational vulnerabilities, delivering strategic recommendations that enhanced system security and process efficiency for enterprise-scale clients.

Consultant, Enterprise IT Risk Management at Deloitte, Hyderabad

JULY 2013 — JULY 2014

- Led comprehensive SOX ITGC control testing to evaluate system integrity, ensuring rigorous adherence to financial reporting requirements and regulatory standards.
- Analyzed and interpreted Service Auditor Reports to identify downstream risks and validate the security posture of critical third-party service providers.